

Armed Conflict and Acute Food Insecurity: Evidence from a High-Frequency District-Level Panel¹

Youssef Kiendrebeogo

IMF

Abstract

This paper estimates the causal effects of armed conflict on acute food insecurity using a newly constructed district-level quarterly panel. We link geocoded conflict events from the Armed Conflict Location and Event Data (ACLED) to Integrated Phase Classification (IPC) food insecurity outcomes across 1,159 districts in 22 developing and fragile countries from 2010 to 2025. A dynamic panel specification that conditions on lagged food crisis outcomes and rich sets of fixed effects yields a positive, precisely estimated, and persistent effect of conflict on food crisis risk. Instrumental-variables estimates that exploit within-country conflict diffusion as a source of exogenous variation imply that conflict raises the probability of food crisis by five to seven percentage points per quarter, with long-run effects that substantially exceed the contemporaneous impact. Mechanism analysis identifies higher food prices, reduced agricultural production, and forced displacement as the principal transmission channels.

Keywords: food insecurity; armed conflict; dynamic panel; instrumental variables; fragile and conflict-affected states; IPC

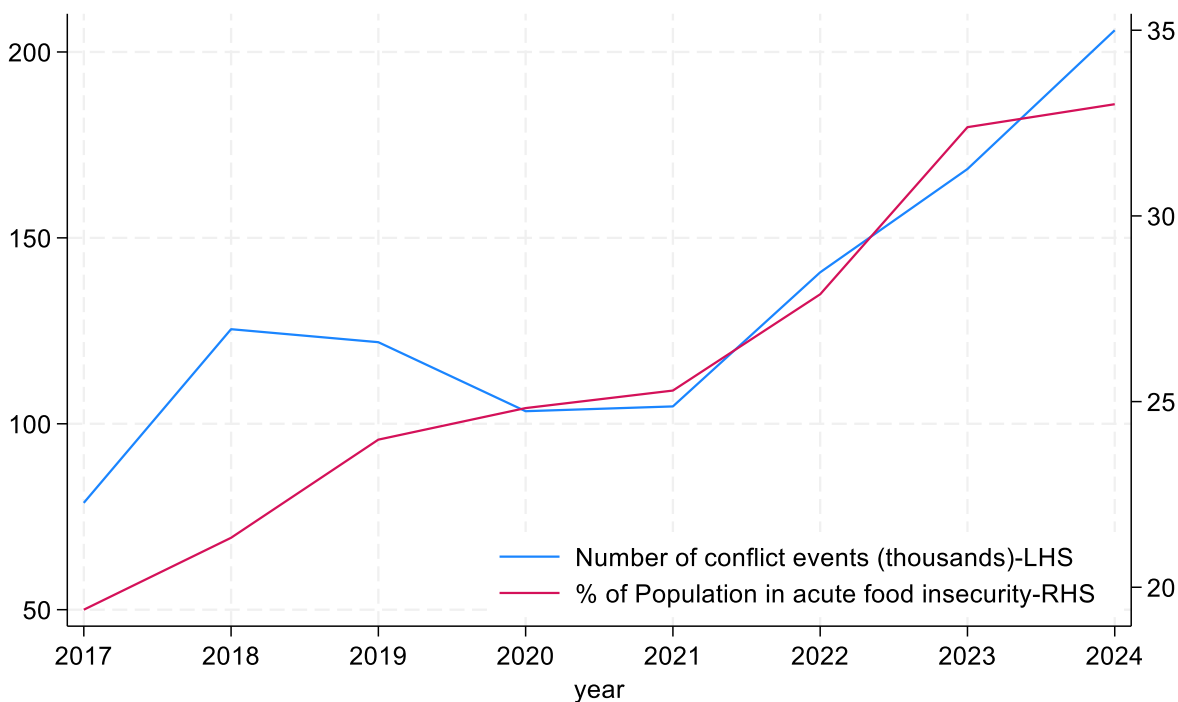
JEL Classification: C23, D74, I15, O12, Q18

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1. Introduction

Acute food insecurity has become a central concern in global policy debates and a defining challenge for the United Nations Sustainable Development Goals, in particular SDG 1 (No Poverty), SDG 2 (Zero Hunger), and SDG 3 (Good Health and Well-Being). Despite sustained international attention and substantial increases in humanitarian expenditure, the burden of acute food insecurity has risen sharply in recent years, particularly in fragile and conflict-affected settings. Recent estimates indicate that more than 343 million people across 74 countries faced acute food insecurity in 2024—approximately 10 million more than in 2023 and nearly 200 million more than before the COVID-19 pandemic (FAO et al., 2024). Particularly alarming is the expansion of catastrophic hunger: roughly 44.4 million people were in emergency conditions or worse, including approximately 1.9 million at risk of famine and requiring immediate life-saving assistance.

Figure 1: Conflict events and prevalence of food insecurity in FCS



Note: The figure plots the number of ACLED conflict events (left axis) and the share of the district-level population classified as IPC Phase 3 or worse (right axis), aggregated annually across all countries in the sample. Source: ACLED; FEWS NET IPC. Source: ACLED and ACLED; FEWS NET IPC.

Figure 1 provides an initial illustration of the relationship between armed conflict and food insecurity in fragile and conflict-affected states. The figure plots the evolution of conflict events and the share of the population experiencing acute food insecurity over time. Two patterns stand out. First, periods of rising conflict tend to coincide with sharp increases in the prevalence of acute food insecurity. Second, this co-movement is especially pronounced in recent years, when both conflict intensity and food insecurity reached historically elevated levels. While this correlation does not establish causation, it underscores a systematic association and motivates an empirical framework capable of isolating causal effects, modeling dynamics, and identifying transmission channels.

A central challenge is measurement. Both conflict and food insecurity are inherently local and evolve rapidly, yet much of the existing cross-country evidence relies on coarse spatial units and low-frequency outcomes that obscure timing, geography, and the persistence of shocks. Conflict exposure is commonly measured at the national level, and food insecurity outcomes are observed infrequently or aggregated across large administrative units. These limitations make it difficult to distinguish contemporaneous from lagged effects, to study persistence, or to trace how conflict propagates through prices, production, and displacement. A parallel strand of the literature uses household- or individual-level microdata within single-country or small-sample settings (Verwimp et al., 2019). While this work provides important causal evidence, its external validity is inherently limited: results from one conflict context may not generalize to countries or periods that differ in institutions, geography, or the nature of violence.

This paper addresses these limitations by constructing a new high-frequency district-level panel that links local conflict exposure to acute food insecurity across countries. We harmonize quarterly data on conflict events from the Armed Conflict Location and Event Data (ACLED) with district-level food insecurity indicators based on the Integrated Phase Classification (IPC), compiled by the Famine Early Warning Systems

Network (FEWS NET). The resulting dataset covers 1,159 districts across 22 developing and fragile countries over 2010–2025.

The construction of this dataset is a core contribution of the paper. ACLED provides geocoded information on the timing, location, and nature of conflict events, which we map to administrative districts and aggregate to the quarterly frequency. We define a binary indicator for food crisis (IPC Phase 3 or worse) at the district-quarter level and align it with conflict exposure, enabling analysis of how local conflict shocks translate into food crises over time.

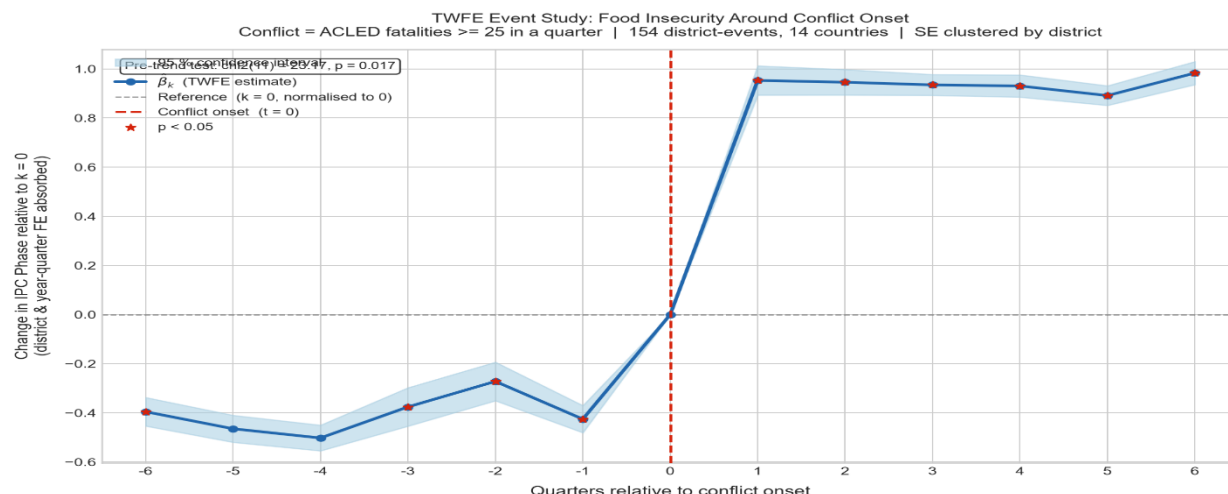
The motivation for building this panel is not purely descriptive. Food insecurity is highly persistent and responds to shocks through multiple channels—market disruptions, agricultural production losses, and forced displacement—whose cumulative effects can unfold over several years. At the same time, conflict may itself be endogenous to local economic conditions, including food insecurity. Figure 2 illustrates this dynamic: it plots food insecurity around the first lethal conflict event in a district, revealing deteriorating food security before conflict onset (consistent with reverse causality) alongside persistently elevated food insecurity afterward (consistent with lasting causal effects). The district-quarter structure of our data allows us to model food insecurity as a dynamic process, conditioning explicitly on lagged food crisis outcomes while absorbing time-invariant heterogeneity through fixed effects.

Armed conflict has long been recognized as a primary driver of food insecurity. In 2024, approximately 65 percent of acutely food-insecure individuals lived in fragile and conflict-affected situations, and every country classified by the World Bank as fragile and conflict-affected recorded a food security rating of “in crisis” or worse. Conflict incidence has itself reached unprecedented levels: the Uppsala Conflict Data Program records 59 state-based conflicts in 2023, the highest since systematic data collection began in 1946, with conflict-related fatalities at their highest level in two decades.

Conflict undermines both the availability of food and households’ access to it. It destroys human and physical capital, disrupts supply chains, reduces agricultural production, and induces large-scale displacement. Communities in conflict-exposed areas commonly face restricted mobility, market closures, constrained access to food assistance, and elevated prices. These mechanisms motivate an empirical analysis that

explicitly models dynamics and transmission channels rather than relying on contemporaneous associations.

Figure 2: Food insecurity before and after conflict onset



Note: The figure displays event-study estimates from a two-way fixed-effects specification of the effect of conflict onset on district-level food insecurity. The event window spans six quarters before and after a district's first lethal conflict event ($k = 0$). The omitted category is the quarter immediately preceding conflict onset. The comparison group consists of districts that never experience a lethal conflict event during the sample period. Standard errors are clustered at the district level. Source: FEWS NET IPC; ACLED.

Using this district-quarter panel, we estimate the short- and long-run effects of conflict on food crisis risk in a dynamic framework that includes lagged food crisis outcomes and time-varying covariates. We complement the baseline estimates with an instrumental-variables strategy exploiting within-country conflict diffusion—the tendency for conflict to spread in waves across districts within a country (Wimmer, 2013; Lake and Rothchild, 1998)—as a source of exogenous variation in local conflict exposure.

The main findings are as follows. First, conflict significantly and persistently raises the likelihood of food crisis. In our preferred IV specifications, the contemporaneous effect is five to seven percentage points per quarter, with long-run effects roughly twice as large. Second, these findings are robust to alternative dynamic specifications and to two complementary IV estimators. Third, the mechanism analysis shows that conflict affects

food insecurity through higher food prices, reduced agricultural production, and population displacement.

The paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data. Section 4 presents the identification strategy. Section 5 reports the main results. Section 6 examines transmission mechanisms. Section 7 concludes.

2. Related Literature

A longstanding question in both the academic and policy literature concerns whether reductions in armed conflict improve food security outcomes. This section surveys the empirical literature on the conflict–food insecurity nexus, organized around three themes: measurement and data limitations, identification challenges, and mechanisms.

Progress in this literature has been constrained by data gaps. Brück and d’Errico (2019) emphasize the need for more granular and timely data to advance both analytical understanding and policy relevance, pointing to the diversity of conflict types and food security outcomes across settings. Shemyakina (2022) similarly stresses the difficulty of drawing general conclusions without data that jointly capture conflict exposure and food security outcomes at sufficiently fine spatial and temporal scales. Our contribution responds directly to these concerns: by constructing a harmonized district-by-country-by-quarter dataset linking ACLED conflict events to IPC-based food insecurity outcomes, we address a key gap identified in the literature.

A substantial body of work has examined the causal relationship between conflict and food insecurity. Martin-Shields and Stojetz (2019) discuss the bidirectional nature of this relationship, emphasizing how food insecurity may itself contribute to the outbreak of violence. Hendrix and Brinkman (2013) review the empirical evidence and highlight the difficulty of isolating causal effects in the presence of endogeneity and omitted variables. Related work documents adverse consequences of conflict for agricultural production, attributing lower output in conflict-affected areas to asset destruction, labor disruptions, reduced access to land and credit, and insecurity (Blattman and Miguel,

2010; Nillesen and Verwimp, 2010; Verpoorten, 2009; Munoz-Mora, 2016; Rockmore, 2015).

The channels through which conflict affects food security remain actively debated. Some studies emphasize state fragility, institutional capacity, and forced displacement as key mediating factors (Verwimp and Munoz-Mora, 2013), while others focus on food prices, market access, and the nature of violence—particularly the intensity of one-sided attacks on civilians (Serneels and Verpoorten, 2015; Eck and Hultman, 2007). Disentangling these mechanisms empirically is difficult when datasets do not allow researchers to track both conflict exposure and intermediate outcomes at high frequency within the same local units.

Most existing empirical studies rely on microdata and employ difference-in-differences or instrumental-variables strategies within single-country settings. Prominent examples include studies of Rwanda (Akresh et al., 2011; Serneels and Verpoorten, 2015), Burundi (Verwimp and Munoz-Mora, 2013), Afghanistan (D’Souza and Jolliffe, 2013), and Côte d’Ivoire (Dabalen and Paul, 2014). This literature has yielded important insights, including evidence that conflict disproportionately harms vulnerable groups such as children, women, and the elderly, with particularly severe effects on nutritional outcomes (Guerrero-Serdan, 2009; Tranchant et al., 2014). Evidence from the Israeli-Palestinian conflict documents a causal chain from violence to reduced income stability and deteriorating food security, though with heterogeneous household responses that may not generalize (Brück and d’Errico, 2019; Arias et al., 2019).

A well-recognized limitation of this micro-level literature is external validity: results from specific conflict settings may not generalize to countries or periods differing in institutions, geography, or the form of violence. Moreover, many studies rely on low-frequency or episodic measures, limiting their ability to study dynamics and persistence (Verwimp et al., 2019).

This paper bridges the gap between detailed local analysis and cross-country generalizability. By constructing a district-level quarterly panel harmonizing geocoded conflict events with IPC-based food insecurity measures across 22 developing and fragile countries, we allow for dynamic analysis while retaining local variation in conflict exposure. The high-frequency structure enables us to distinguish short-run from long-

run effects, model food insecurity as a persistent process, and examine transmission mechanisms within a single unified empirical framework—responding directly to the data and identification challenges highlighted in the literature.

3. Data and Measurement

Our analysis rests on a harmonized district-by-country-by-quarter panel that links geocoded conflict events to high-frequency measures of acute food insecurity. This structure aligns the timing of conflict shocks with transitions into food crisis conditions and allows us to model food insecurity as a dynamic process rather than a sequence of independent episodes.

3.1. Food Insecurity

The outcome variable, $FoodCrisis_{dct}$, is a binary indicator equal to one if district d in country c is in food crisis in quarter t , and zero otherwise. Food crisis is defined using the Integrated Phase Classification (IPC) system, which categorizes acute food insecurity into five phases: Phase 1 (Minimal), Phase 2 (Stressed), Phase 3 (Crisis), Phase 4 (Emergency), and Phase 5 (Famine). IPC classifications are compiled at the monthly frequency by the Famine Early Warning Systems Network (FEWS NET) for a broad set of fragile and conflict-affected countries.

Monthly IPC data are collapsed to the quarterly frequency using an “any-month” rule: a district is coded as food-insecure in quarter t if it records IPC Phase 3 or worse in at least one month of that quarter. This rule appropriately weights short-lived but severe deteriorations, which are operationally salient for humanitarian response and early warning. A strict majority or averaging rule would attenuate precisely those acute episodes that motivate the analysis.

3.2. Conflict Exposure

For conflict exposure, we draw on event-level data from the Armed Conflict Location and Event Data (ACLED; Raleigh et al., 2010), which provides geocoded information on the timing and location of conflict events and captures forms of political violence beyond conventional battle deaths. Geographic granularity is particularly valuable in the food security context, where violence in one location can generate spillovers through market access, mobility constraints, and supply chain disruptions.

The binary conflict indicator $Conflict_{dct}$ equals one if district d in country c records at least one ACLED event associated with 25 or more fatalities during quarter t , and zero otherwise. The 25-fatality threshold aligns with the conventional UCDP battle-death criterion for state-based armed conflict and is applied at the event level, so that a single severe incident—rather than an accumulation of minor episodes—defines conflict exposure for the district-quarter.

3.3. Additional Covariates

We include a set of time-varying district-level covariates that the empirical literature on food crisis prediction identifies as important determinants of food insecurity (Andrée et al., 2022). These include rainfall anomalies, evapotranspiration, population size, staple food price inflation, the Rural Access Index, terrain ruggedness (following Nunn and Puga, 2012), the Normalized Difference Vegetation Index (NDVI), and cropland and pasture area shares.

4. Identification Strategy

4.1. Baseline Dynamic Specification

Estimating the effect of conflict on food insecurity is complicated by persistence in food crises and the potential endogeneity of conflict incidence. We address both issues with a dynamic panel specification that conditions on lagged food crisis outcomes and a rich set of fixed effects.

The preferred specification is:

$$\text{FoodCrisis}_{dct} = \sum_{k=1}^p \delta_k \text{FoodCrisis}_{dc,t-k} + \beta \text{Conflict}_{dct} + \theta' X_{dct} + \gamma_d + \gamma_c + \gamma_t + \varepsilon_{dct} \quad (1)$$

where γ_c , γ_d and γ_t denote country, district and year-quarter fixed effects; X_{dct} is a vector of time-varying covariates; and $K \leq 3$ is the number of lagged food crisis terms. The coefficient of interest, β , captures the change in the probability of food crisis associated with severe conflict exposure, holding constant lagged food insecurity dynamics, observed covariates, and fixed effects.

Identification rests on sequential exogeneity: conditional on fixed effects, observed covariates, and lagged food crisis outcomes, current shocks to food insecurity are orthogonal to current conflict exposure. Districts with conflict are assumed not to follow systematically different underlying food security trends once we condition on past food insecurity and absorb time-invariant heterogeneity through fixed effects.

The coefficient β captures the contemporaneous impact of conflict. The long-run multiplier, which accounts for dynamic feedback through the lag coefficients, is:

$$LR = \beta / (1 - \sum_{k=1}^p \delta_k) \quad (2)$$

This long-run effect captures the cumulative impact of a permanent conflict shock once the food insecurity process has fully adjusted. We report long-run estimates alongside contemporaneous effects throughout.

We estimate equation (1) using both a linear probability model (LPM) and a probit specification. In the probit columns, coefficients are reported as average marginal effects evaluated at sample means. The sample covers 1,159 districts in 22 developing and fragile countries, spanning 2010–2025, with sample size varying across specifications as more lags require longer consecutive observation histories.

4.2. Instrumental Variables Strategy

The baseline specification addresses persistence and time-invariant heterogeneity but may leave residual endogeneity if time-varying shocks—such as local economic disruptions, political instability, or measurement error in conflict exposure—jointly affect conflict incidence and food insecurity. We therefore complement the baseline estimates with an instrumental-variables strategy.

Instrument construction. Our IV strategy exploits the empirical regularity that conflict tends to spread in waves within countries (Wimmer, 2013; Lake and Rothchild, 1998): a district’s exposure to conflict is strongly related to contemporaneous conflict incidence in other districts within the same country. We use this within-country diffusion of conflict as a source of variation in local conflict exposure that is plausibly orthogonal to district-level food insecurity innovations, once we condition on lagged food insecurity and fixed effects.

Formally, let t_0 denote the initial period of the sample and let C_{dct_0} indicate whether district d in country c experienced conflict in t_0 . Define the set of districts in the same country that share the same initial conflict status as district d by

$$\mathcal{J}_{dc} \equiv \{d' \neq d: S_{d'} = S_d, C_{d'ct_0} = C_{dct_0}\},$$

where S_d denotes the country containing district d . The “conflict-wave” instrument Z_{dct} is the leave-one-out average conflict incidence among districts in $\mathcal{J}_{dc}$ at time t :

$$Z_{dct} \equiv \frac{1}{|\mathcal{J}_{dc}|} \sum_{d' \in \mathcal{J}_{dc}} C_{d'ct} \quad (3)$$

By construction, Z_{dct} captures shifts in conflict exposure that are common within a country and initial-status group, while excluding the district's own contemporaneous conflict realization. We use the first lag, $Z_{dc(t-1)}$, as the excluded instrument, providing an additional time lag that further attenuates the risk of reverse causality.

Exclusion restrictions: The identifying assumption is that, conditional on lagged food insecurity, observed covariates, and district and year-quarter fixed effects, conflict waves in other districts of the same country affect district i 's food insecurity only through their impact on district d 's conflict incidence. While national conflict waves could in principle affect food insecurity through aggregate channels, the inclusion of year-quarter fixed effects and lagged food crisis dynamics absorbs such common shocks.

Estimators. We implement two complementary IV estimators suited to the binary outcome. First, we apply the IV-Probit control function approach (Rivers and Vuong, 1988), which augments the second-stage probit with first-stage residuals. Second, we use the special regressor (SR) estimator of Lewbel (2000) and Dong and Lewbel (2015), a semiparametric alternative that accommodates limited dependent variables with endogenous regressors under weaker distributional assumptions. The SR estimator avoids iterative maximum-likelihood optimization, permits heteroskedasticity of unknown form, and does not require correct specification of the first-stage equation for consistency. We treat the SR estimates as our preferred IV results.

5. Results

5.1. Baseline Estimates

Table 1 reports baseline estimates of equation (1) using both the LPM (Columns 1–4) and probit (Columns 5–8). Across all specifications, the estimated effect of conflict on food crisis risk is positive and statistically significant at conventional levels.

In the LPM, the marginal effect of conflict exposure ranges from 0.013 (Column 3) to 0.029 (Column 1) across specifications. Probit average marginal effects range from 0.017 to 0.038. The stability of these estimates across functional forms indicates that the main findings are not driven by the choice of link function.

Table 1: Marginal response of food insecurity

	Linear Probability Model				Probit Model			
Regressors	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Conflict	0.029*** (0.008)	0.018*** (0.005)	0.013*** (0.003)	0.024*** (0.006)	0.023*** (0.006)	0.038** (0.011)	0.017*** (0.004)	0.021*** (0.006)
Food Crisis, first lag	0.013** (0.005)	0.011** (0.004)	0.011** (0.004)	0.010** (0.004)	0.018* (0.008)	0.017** (0.007)	0.013* (0.005)	0.014* (0.006)
Food Crisis, 2nd lag		-0.072 (0.051)	-0.050 (0.041)	-0.014* (0.008)		-0.052 (0.040)	-0.075 (0.062)	-0.011* (0.006)
Food Crisis, 3rd lag			0.092* (0.053)	0.063* (0.036)			0.105* (0.060)	0.097** (0.040)
p-value, lags 4-8				0.135				0.118
Long-run impact of Conflict	0.082** (0.035)	0.104** (0.043)	0.110** (0.047)	0.104*** (0.031)	0.093** (0.040)	0.105** (0.043)	0.091*** (0.027)	0.066*** (0.019)
Rainfall				-0.128** (0.055)				-0.113** (0.049)
District Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SE clustered by district	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	37,351	31,132	26,042	26,036	37,346	31,126	26,036	26,036
No. of districts	1,159	1,159	1,112	1,112	1,159	1,159	1,112	1,112
No. Of countries	22	22	20	20	22	22	20	20
R2	0.524	0.498	0.459	0.459	-	-	-	-
Pseudo R2	-	-	-	-	0.482	0.471	0.452	0.452

Note: Dependent variable is a binary indicator equal to 1 if the district is in IPC food crisis Phase ≥ 3 in the quarter. Conflict_dct = 1 if ACLED-recorded fatalities ≥ 25 in the district-quarter. Columns 1–4 are estimated by OLS with district and year-quarter fixed effects; columns 5–8 are probit models with country and year-quarter fixed effects, reporting average marginal effects evaluated at the sample means. The long-run impact equals $\beta_e / (1 - \sum \gamma_k)$, where β_e is the conflict coefficient and γ_k are the lag coefficients (or

their marginal-effect analogues for probit). Standard errors in parentheses are clustered at the district level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Data: FEWS NET IPC (quarterly), ACLED conflict events.

Food insecurity is highly persistent. The first lag of the food crisis indicator enters positively and significantly in all specifications (e.g., 0.013 in Column 1; 0.014 in Column 8), reflecting strong state dependence in acute food insecurity. The dynamic pattern is not purely autoregressive: when additional lags are included, the second lag is negative and imprecisely estimated in several specifications (e.g., -0.072 in Column 2; -0.075 in Column 7), while the third lag is positive and statistically significant (e.g., 0.092 in Column 3; 0.097 in Column 8). The joint test for higher-order lags (4–8) yields p -values of 0.135 and 0.118 in Columns 4 and 8, respectively, indicating that the predictive content of past crises is well captured by the first three lags.

Columns 4 and 8 add rainfall as an additional time-varying control. Rainfall enters with a negative and statistically significant coefficient (-0.128 in Column 4; -0.113 in Column 8), consistent with the role of adverse agroclimatic conditions in elevating food crisis risk. Controlling for rainfall leaves the conflict coefficient essentially unchanged (0.024 in Column 4 versus 0.013 in Column 3; 0.021 in Column 8 versus 0.017 in Column 7), reinforcing the robustness of the conflict–food insecurity relationship.

The implied long-run impact of conflict is economically meaningful and broadly consistent across functional forms. In the most saturated specifications (Columns 4 and 8), the long-run multipliers are 0.104 and 0.066, respectively. While magnitudes vary with the lag structure—as expected in dynamic models—the presence of a positive long-run effect across specifications is consistent with the view that conflict elevates food crisis risk well beyond the contemporaneous quarter.

5.2. Instrumental Variables Estimates

Table 2 reports IV estimates of equation (1) using the SR estimator (Columns 1–4) and IV-Probit (Columns 5–8), with coefficients expressed as marginal effects. Across all specifications, the IV estimates are positive, precisely estimated, and statistically significant.

Table 2: Marginal response of food insecurity (IVM, second stage)

	SR Model (kernel density)				IV Probit Model			
Regressors	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Conflict	0.073*** (0.020)	0.058*** (0.017)	0.064*** (0.019)	0.052*** (0.012)	0.066*** (0.021)	0.050*** (0.015)	0.048*** (0.013)	0.071*** (0.021)
Food Crisis, first lag	0.027*** (0.007)	0.020*** (0.006)	0.019*** (0.005)	0.026*** (0.008)	0.029*** (0.008)	0.026*** (0.005)	0.024*** (0.007)	0.029*** (0.008)
Food Crisis, 2nd lag		-0.083 (0.090)	-0.071 (0.095)	-0.047 (0.083)		-0.082 (0.090)	-0.085 (0.082)	-0.071 (0.086)
Food Crisis, 3rd lag			0.084** (0.036)	0.071** (0.032)			0.094** (0.040)	0.092** (0.041)
p-value, lags 4-8				0.119				0.105
Long-run impact of Conflict	0.129*** (0.037)	0.131*** (0.039)	0.128*** (0.041)	0.130*** (0.036)	0.123*** (0.037)	0.136*** (0.041)	0.125*** (0.033)	0.137*** (0.039)
Rainfall				-0.093* (0.053)				-0.108* (0.062)
District Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SE clustered by district	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	37,245	31,040	25,962	25,962	37,245	31,040	25,962	25,962
No. of districts	1,156	1,156	1,109	1,109	1,156	1,156	1,109	1,109
No. of countries	22	22	20	20	22	22	20	20
Hansen J statistic (p-value)	0.203	0.211	0.195	0.208	-	-	-	-
Pseudo R2	-	-	-	-	0.219	0.209	0.218	0.217

Note: Dependent variable = 1 if IPC Phase ≥ 3 . Conflict = 1 if ACLED fatalities ≥ 25 . Cols 1-4: SR (Lewbel 2000) estimator with special regressor $V = \text{Rainfall}$; $W = (Y - G(V))/g(V)$; marginal effects = $\beta + \text{mean}(g_V)$. Cols 5-8: IV-Probit control function (Rivers-Vuong); marginal effects at sample means. Instrument: first lag of leave-one-out average conflict, same country x initial-status group. Long-run = $\beta_{\text{conflict}}/(1 - \sum(\gamma_{\text{lags}}))$. SE clustered by district. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The contemporaneous marginal effect of conflict ranges from 0.052 to 0.073 in the SR columns and from 0.048 to 0.071 in the IV-Probit columns, implying a 4.8–7.3 percentage point increase in the probability of food crisis in a conflict-exposed district-

quarter. These IV magnitudes exceed the corresponding baseline estimates (0.013–0.029 in LPM; 0.017–0.038 in probit), consistent with attenuation in the baseline due to measurement error in conflict exposure or residual endogeneity bias. The implied long-run effects are also economically substantial in the most saturated specifications: 0.130 in SR Column 4 and 0.137 in IV-Probit Column 8.

Table 3: Marginal response of food insecurity (IVM, first stage)

	SR Model (kernel density)				IV Probit Model			
Regressors	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Conflict waves	0.104*** (0.029)	0.105*** (0.027)	0.104*** (0.031)	0.108*** (0.027)	0.093*** (0.025)	0.098** (0.027)	0.097*** (0.023)	0.091*** (0.025)
Food Crisis, first lag	0.026*** (0.007)	0.021*** (0.006)	0.021*** (0.005)	0.020** (0.008)	0.035*** (0.010)	0.030** (0.012)	0.028*** (0.008)	0.039*** (0.011)
Food Crisis, 2nd lag		-0.101 (0.162)	-0.105 (0.153)	-0.093 (0.128)		-0.083 (0.106)	-0.099 (0.102)	-0.092 (0.106)
Food Crisis, 3rd lag			0.050** (0.021)	0.053** (0.024)			0.083** (0.036)	0.082** (0.037)
Rainfall				-0.114 (0.142)				-0.103 (0.095)
District Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SE clustered by district	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	37,245	31,040	25,962	25,962	37,245	31,040	25,962	25,962
No. of districts	1,156	1,156	1,109	1,109	1,156	1,156	1,109	1,109
No. of countries	22	22	20	20	22	22	20	20
R ²	0.328	0.429	0.358	0.349	0.224	0.291	0.240	0.247
Excl. IV F-statistics	118.4	106.2	116.8	126.7	129.3	141.5	128.4	144.1

Note: Dependent variable = Conflict_dct (1 if ACLED fatalities $\geq$ 25). Instrument (Z_{t-1}): first lag of leave-one-out average conflict incidence among districts in the same country x initial-conflict-status group. All specifications include district and year-quarter fixed effects. SE clustered by district. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3 reports first-stage results. The conflict-wave instrument is strongly predictive of district-level conflict exposure across all specifications, with excluded-instrument F-statistics ranging from approximately 106 to 144. These values substantially exceed conventional weak-instrument thresholds, supporting the credibility of the IV

estimates. The consistency of findings across the SR and IV-Probit estimators—which rely on different identifying assumptions and computational strategies—reinforces the conclusion that conflict exerts a genuine causal effect on food crisis risk.

6. Mechanisms

6.1. Empirical Framework

This section examines the channels through which conflict affects acute food insecurity. The baseline specification in equation (1) deliberately excludes several economic and institutional variables that may lie on the causal pathway from conflict to food crises. We treat these variables as candidate intermediating outcomes: their responses to conflict are informative about mechanisms, even though they may also respond to other shocks or feed back into conflict and food insecurity.

For each candidate mechanism M_{dct} , we estimate a dynamic panel specification that mirrors equation (1):

$$M_{dct} = \sum_{j=1}^p \delta_j Y_{dc,t-j} + \sum_{j=1}^p \phi_j M_{dc,t-j} + \beta C_{dct} + X'_{dct} \theta + \gamma_d + \gamma_c + \gamma_t + \varepsilon_{dct} \quad (4)$$

Lags of both the mechanism and the food crisis indicator control for persistence and potential reverse causality. As in Section 4.2, we instrument conflict with the country-level conflict wave and estimate equation (4) by two-stage least squares. We report both contemporaneous and long-run effects for each mechanism.

6.2. Channels and Data

We examine four candidate channels: food prices, food production, population density (a proxy for displacement and demographic sorting), and sub-national human development. For mechanisms 1 through 3, estimation uses the full district-quarter panel. The Human Development Index (HDI) is available only at the country level from the UNDP Human Development Reports; the HDI regressions therefore use a substantially smaller sample (202 country-year observations from 22 countries), which limits statistical power and comparability with the other mechanism estimates.

6.3. Results

Table 4 reports estimates from equation (4) for all four mechanisms, presenting baseline OLS (Columns 1–4) and two-stage least squares IV estimates (Columns 5–8).

Table 4: Potential channels of transmission (IVM, second stage)

Regressors	Food prices	Food production	Population density	Subnational HDI
Conflict	0.083*** (0.025)	-0.128** (0.055)	-0.092*** (0.038)	-0.038 (0.042)
Conflict, first lag	0.104*** (0.029)	0.105*** (0.027)	0.104*** (0.031)	0.108*** (0.027)
Conflict, 2nd lag	0.026*** (0.007)	0.021*** (0.006)	0.021*** (0.005)	0.020** (0.008)
Conflict, 3rd lag	0.114 (0.142)	-0.101 (0.162)	-0.105 (0.153)	-0.093 (0.128)
Rainfall	0.104*** (0.031)	0.020** (0.008)	0.050** (0.021)	0.053** (0.024)
Long-run impact of conflict	0.113*** (0.034)	-0.115*** (0.031)	-0.104* (0.059)	-0.083 (0.119)
District Fixed Effects	Yes	Yes	Yes	No
Country Fixed Effects	Yes	Yes	Yes	Yes
Year-quarter Fixed Effects	Yes	Yes	Yes	No
Year Fixed Effect	No	No	No	Yes
SE clustered by district	Yes	Yes	Yes	Yes
Observations	30,244	30,161	30,652	202
R2	0.328	0.429	0.358	0.349
Excl. IV F-statistics (first stage)	118.4	106.2	116.8	126.7

Note: Cols 1–4: OLS (baseline). Cols 5–8: 2SLS with excluded instrument = lagged conflict wave Z_{t-1} . Mechanisms: (1)/(5) Food Prices = $\log(\text{staple food price, USD/kg})$; (2)/(6) Food Production = NDVI mean (vegetation/crop proxy); (3)/(7) Population Density = $\log(\text{population} / \text{area})$; (4)/(8) HDI = Human Development Index (UNDP, 0–1 scale, country-year panel). Specification: 3 lags of mechanism + 3 lags of food crisis indicator. SE clustered by district (country for HDI). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Food prices. Conflict is associated with a statistically significant increase in food prices in both specifications. The contemporaneous coefficient is 0.077 (s.e. 0.024) in the OLS column and 0.083 (s.e. 0.025) in the IV column, both significant at the 1 percent level. The implied long-run impacts are also positive and statistically significant: 0.102 (s.e. 0.044) in OLS and 0.113 (s.e. 0.034) in the IV column. These results indicate that conflict generates persistent upward pressure on staple food prices, consistent with disruptions to market access, distribution networks, and local infrastructure that primarily operate through the food access channel.

Food production. Conflict is associated with a significant decline in food production. The contemporaneous OLS estimate is -0.120 (s.e. 0.031), significant at the 1 percent level; the IV estimate is -0.128 (s.e. 0.055), significant at the 5 percent level. Implied long-run effects are negative and economically meaningful (-0.166 , s.e. 0.075, in OLS; significantly negative in IV). These patterns are consistent with conflict reducing local food availability through curtailed cultivation, labor reallocation away from agriculture, disrupted input supply, and physical damage to productive assets.

Population density. Conflict is associated with a significant decline in population density, consistent with displacement from conflict-affected areas. The OLS estimate is -0.094 (s.e. 0.028), significant at the 1 percent level; the IV estimate is -0.092 (s.e. 0.038), significant at the 5 percent level. Long-run effects are also negative (-0.115 , s.e. 0.048, in OLS; -0.104 , s.e. 0.059, in IV), indicating that conflict induces not only short-run mobility responses but also protracted displacement with slow return dynamics. This channel

affects food insecurity both directly—through loss of agricultural livelihoods—and indirectly, through disruptions to local production and market activity.

Human development index. For HDI, contemporaneous coefficients are negative in both models (-0.029 in OLS; -0.038 in IV) but statistically indistinguishable from zero. Long-run effects are also negative (-0.083 in both specifications) and imprecisely estimated. The limited statistical power reflects the small sample and high aggregation level: with only 202 country-year observations, within-unit variation is insufficient to identify a precise effect in this specification. The negative point estimates are consistent with adverse welfare effects of conflict on human development, but the HDI results should be interpreted with caution.

6.4. Summary

The mechanism estimates collectively indicate that conflict affects food crisis risk through three reinforcing channels: higher food prices, reduced agricultural production, and population displacement. These correspond to disruptions in food access, food availability, and livelihoods, respectively. All three channels are statistically and economically significant both contemporaneously and over the long run and are robust to the IV correction for endogeneity. Together, they provide a coherent explanation for the magnitude and persistence of the long-run conflict effects documented in Section 5, and align closely with the channels emphasized in the conceptual literature on conflict and food security.

7. Concluding Remarks

This paper estimates the short- and long-run effects of armed conflict on acute food insecurity using a newly constructed, high-frequency district-level panel spanning

22 developing and fragile countries. By explicitly modeling the persistence of food insecurity and exploiting within-district variation in conflict exposure, the analysis establishes that conflict substantially raises the probability of food crisis both contemporaneously and over time. Baseline linear probability and probit specifications yield a positive, statistically significant, and robust association between conflict exposure and food crisis risk even after conditioning on lagged food insecurity and rich sets of fixed effects.

The complementary instrumental-variables strategy, exploiting within-country conflict diffusion as a source of exogenous variation in district-level conflict incidence, corroborates the baseline findings while yielding larger estimates consistent with attenuation bias in the baseline. Conflict increases the probability of food crisis by five to seven percentage points per quarter, with implied long-run effects roughly twice as large. Strong first-stage relationships and large excluded-instrument F-statistics support instrument relevance, and the consistency of results across two complementary IV estimators reinforces the interpretation of a genuine causal effect.

Beyond establishing causality, the paper documents the pathways through which conflict propagates into food crises. Using the same dynamic panel framework, the mechanism analysis shows that conflict leads to sustained increases in food prices, declines in local agricultural production, and reductions in population density consistent with forced displacement. These channels map naturally to disruptions in food access, availability, and livelihoods, and together provide a coherent account of the magnitude and persistence of the long-run conflict effects.

The paper makes three broader contributions. First, it introduces a harmonized district-by-country-by-quarter dataset linking geocoded conflict events to IPC-based food insecurity measures across a large cross-section of developing and fragile countries, directly addressing measurement gaps identified in the literature. Second, it provides

dynamic estimates of both short- and long-run effects and advances causal identification through an instrument grounded in observed patterns of conflict diffusion. Third, it offers systematic evidence on mechanisms within a single empirical framework, helping to bridge reduced-form estimates and the channels emphasized in humanitarian policy discussions.

These findings carry direct policy relevance. The food insecurity consequences of conflict are not confined to the immediate aftermath of violence: they persist through self-reinforcing mechanisms operating over time. Effective policy responses therefore need to combine short-term humanitarian relief with measures to stabilize food prices, protect agricultural production, and support displaced populations. More broadly, the results underscore the value of high-frequency, spatially disaggregated data for improving early-warning systems and for designing interventions aimed at mitigating the long-run food security consequences of armed conflict.

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